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MILITARY STANDARD INSPECTION PROCEDURE FOR DETERMINING THE MAGNETIC PERMEABILITY OF WROUGHT AUSTENITIC STEELS

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OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE
WASHINGTON 25, D. C.

26 JULY 1956

Supply and Logistics

Inspection Procedure for Determining the Magnetic Permeability of Wrought Austenitic Steels

MIL–STD–288

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force effective upon publication of this document.

2. In accordance with established procedure, the Standardization Division has designated the Ordnance Corps, Bureau of Ships, and Air Force, respectively, as Army-Navy-Air Force custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Office of the Assistant Secretary of Defense (Supply and Logistics), Washington 25, D. C.

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MILITARY STANDARD

INSPECTION PROCEDURE FOR DETERMINING THE MAGNETIC PERMEABILITY OF WROUGHT AUSTENITIC STEELS

1. SCOPE

1.1 This standard covers the inspection procedure which shall be used for determining the magnetic permeability of wrought austenitic steel applicable to purchases where essentially nonmagnetic material is specified. This standard shall not be used for material intended for periscope tubes or accessories.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, of the issue in effect on date of invitation for bids, form a part of this standard:

SPECIFICATIONS

FEDERAL

QQ–M–151 — Metals, General Specifications for Inspection of.

QQ–S–763 — Steel Bars, Shapes, and Forgings — Corrosion Resisting.

QQ–S–766 — Steel, Corrosion Resisting: Plates, Sheets and Strips.

MILITARY

MIL–S–854 — Steel, Corrosion-Resisting: Plates, Sheets, Strips, and Structural Shapes.

MIL–I–17214 — Indicator, Permeability; Low MU (Go-No-Go).

MIL–S–17758 — Steel Bar and Steel Forging, Nickel Modified Hadfield and Nickel-Molybdenum Modified Hadfield.

MIL–S–17759 — Steel Bar, Billet, Forging and Wire, Chromium-Nickel-Phosphorous Alloy, Low Magnetic Permeability.

MIL–S–17996 — Steel Sheet and Other Wrought Forms, Corrosion Resisting, Austenitic (Low Magnetic Permeability), Chromium-Manganese-Nickel.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials

3.1.1. The inspection procedures detailed herein are applicable to the austenitic steels. Separate procedures are specified, depending upon the class of materials (see 3.1.2.1).

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3.1.2. *Classes of materials.*–

3.1.2.1. For inspection purposes wrought austenitic steels are divided into the following classes:

Class 1 — Those analyses that are not known to exhibit significant tendencies toward ferrite segregation.

Class 2 — Those analyses that are known to exhibit significant tendencies toward ferrite segregation.

3.1.2.2 *Class 1 material.*—Class 1 material includes the compositions that conform to the following AISI standard steel types, specifications, or descrpitions:

AISI–302
AISI–303
AISI–304
AISI–305
AISI–310

MIL–S–854
Class 1—condition A
Class 2—condition A
Class 11—condition A

QQ–S–763
Class FS–302, condition A
Class FS–303, condition A
Class FS–304, condition A
Class FS–310, condition A

QQ–S–766
Class FS–302—condition A
Class FS–304—condition A
Class FS–310—condition A

3.1.2.2.1 All manganese type austenitic steels that depend primarily on manganese for austenite stability but which may also contain various amounts of nickel, chromium, and molybdenum, in addition to the elements ordinarily found in carbon steel, shall be in class 1.

MIL–S–17758
Alloy 1
Alloy 2

MIL–S–17996
Class 1
Class 2

3.1.2.3 *Class 2 material.*—Class 2 material includes compositions conforming to the following AISI standard steel types, specifications, or descriptions:

AISI–321
AISI–347
AISI–304L
AISI–316
AISI–316L

MIL–S-854
Class 8
Class 9

QQ–S–763
Class FS-304 L
Class FS-316
Class FS-316 L
Class FS-321
Class FS-347

QQ–S–766, condition A of the following classes:
Class FS-321
Class FS-347
Class FS-304L
Class FS-316
Class FS-316L
MIL–S–17759

3.1.3 Austenitic steels that are not included in class 1 or 2 shall be considered class 2 material.

3.1.4 *Permeability.*—The permeability of classes 1 and 2 shall not exceed 1.2.

3.2 Inspection equipment

3.2.1 A permeability indicator[1] conforming to Specification MIL–I–17214 shall be used

[1] Indicators, permeability; Low Mμ (Go-No-Go) are available to Government inspectors on request to the Chief of the Bureau of Ships, Navy Department, Washington 25, D. C.

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for inspecting class 1 material as specified in 4.2.1.

3.2.2 A permeameter[2] shall be used for inspecting class 2 material as specified in 4.2.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling.

4.1.1 *Lot.*—

4.1.1.1 *Class 1 material furnished in plate form.*—For purposes of sampling, where class 1 material is furnished in plate form, each finished plate shall be considered a lot.

4.1.1.2 *Class 2 material furnished in plate form.*—For purposes of sampling, where class 2 material is furnished in plate form, a lot shall consist of all plates of one thickness from one heat that have been heat-treated at the same time or that have been given the same heat treatment cycle for time and temperature. For continuous heat treatment, a lot shall consist of all plates of the same thickness from one heat that are heat-treated in a continuous period. Where a large plate has been sheared or cut into two or more smaller plates, the material represented by the original large plate shall be considered a plate.

4.1.1.3 *Class 1 or 2 material furnished in other than plate form.*—For purposes of sampling where class 1 or 2 material is furnished in other than plate form, a lot shall consist of all the material of one form from one heat that has been heat-treated at the same time or that has been given the same heat-treatment cycle for time and temperature. For continuous heat treatment, a lot shall consist of all the material of one form from one heat that is heat-treated in a continuous period. For hot rolled material, a lot shall consist of all the material of the same form from a heat.

[2] The permeameter is used as per ASTM Standard Designation A342–49 "Standard Methods of Test for Permeability of Feebly Magnetic Materials."

4.1.1.4 When the melt or heat of either annealed or hot rolled material cannot be identified, a lot shall consist of not more than 20 pieces of the same form, analysis, and condition.

4.1.2 *Sampling procedure.*—

4.1.2.1 *Class 1 material.*—For plate material, each finished plate shall be inspected as specified in 4.2.1. For material in other forms, four pieces shall be selected by or under the supervision of the Government inspector from each lot. Each piece shall consist of any individual coil, sheet, strip, bar, or shape. If the number of pieces in a lot is less than the number of pieces to be inspected, each piece in the lot shall be inspected as specified in 4.2.1.

4.1.2.2 *Class 2 material.*—

4.1.2.2.1 For class 2 material furnished in plate form, one specimen shall be prepared by the contractor from each plate selected by or under the supervision of the Government inspector from each lot (see 4.1.1.2), in accordance with table I, and shall be subjected to the inspection specified in 4.2.2.

TABLE I. *Sampling of class 2 plates*

Number of plates in lot	Number of sample plates required	
	Less than ½ inch thick	½ inch thick or more
1	1	1
2 to 5	2	3
6 to 15	2	5
16 to 25	3	7
26 to 65	4	10
66 to 110	5	15
111 to 180	7	15
181 to 300	10	25
301 to 500	15	35

4.1.2.2.2 For class 2 material furnished in other than plate form, two specimens shall be prepared by the contractor from material selected by or under the supervision of the

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Government inspector from each lot, and shall be subjected to the inspection specified in 4.2.2. Specimens of the following dimensions shall be prepared from each plate or piece:

(1) *Round and hexagon stock.*—Solid cylinders ½ inch diameter by 3 inches long.

(2) *Structural shapes.*—(I beams, angles, tee bars, etc.) Sufficient pieces 1⅜ inches wide by 3 inches long by full web or leg thickness to make a stack ½ inch high. For stock over ½ inch thickness of web or leg the specimens shall be 1⅜ inches wide by 3 inches long by ½ inch thick.

(3) *Flat stock.*—(Plates, sheets, strips, bars, etc.) Sufficient pieces 1⅜ inches wide by 3 inches long by stock thickness to make a stack ½ inch high. For stock over ½ inch thick the specimens shall be 1⅜ inches wide by 3 inches long by ½ inch thick.

4.1.2.2.3 Specimens taken from material greater than ½ inch in thickness or diameter shall be taken in such a manner that the longitudinal axis of the specimen shall include the mid-point of the thickness of the material.

4.1.2.2.4 For class 2 forgings, specimens shall be prepared either from forgings or from the forging stock used in the manufacture of the lot. These specimens shall have gone through all the heating and cooling cycles to which the forgings have been subjected. The specimens shall be ½ inch diameter by 3 inches long or 1⅜ inches wide by 3 inches long by ½ inch thick.

4.2 Inspection.

4.2.1 *Class 1 material.*—Each of the plates or pieces selected as specified in 4.1.2.1 shall be inspected by the Government inspector with the permeability indicator, using the 1.2 insert. The number of indications taken on each piece shall be at least equal to one-fifth the area (one side only) for flat sections in square feet of the piece with a minimum of 3 indications per piece. For example: a 5-foot by 4-foot plate would require at least 4 indications. Indications shall be taken on surfaces ready for shipment. Any indications reflecting the increase in permeability resulting from the cold shearing of stock to size shall not be cause for rejection. All indications shall establish that the permeability of the material conform to 3.1.4.

4.2.2. *Class 2 material.*—Each specimen selected in accordance with 4.1.2.2 shall be sent to the laboratory designated by the contracting officer for permeability tests.

4.2.3 *Rejection.*—

4.2.3.1 *Class 1 material.*—If any portion of the plate or piece fails to conform to 4.2.1, the lot represented by that plate or piece shall be rejected. Material from such lots may be resubmitted after sorting or other treatment by the contractor, subject to acceptance or rejection on the basis of inspections made on each individual piece in the lot by the Government inspector.

4.2.3.2 *Class 2 material.*—If any specimen fails to conform to 4.2.2. the lot represented by that specimen shall be considered rejected. When the permeability does not exceed 2.0, rejected lots may be reconsidered by the bureau or agency concerned, taking into consideration the apparent degree of segregation.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Copies of this standard for military use may be obtained as indicated in the foreword to the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms, and contractors from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

Both the title and identifying symbol number should be stipulated when requesting copies of military standards.

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